INDIAN RAILWAYS TRAIN DELAY PREDICTION & PUNCTUALITY ANALYTICS

Three-Person Live Demonstration & Presentation Speech | Course: ECSCI24302 Machine Learning Essentials

Team Members & Roles	Academic Context
• **Dhruv:** Team Lead — Problem Definition, Dataset Scope & Predictor Demo • **Darshi:** Team Member — Dataset Analysis, Preprocessing & Feature Engineering • **Kirtan:** Team Member — Algorithm Benchmarks, LightGBM Tuning & Error Analysis	• **Institution:** Faculty of Engineering & Technology, Adani University • **Program:** B.Tech Computer Science & Engineering (2nd Year) • **Target Time:** 7 to 9 minutes total (~2.5 to 3.0 min per speaker) • **Champion Model:** Tuned LightGBM (Accuracy: 85.85%, Recall: 92.10%, F1: 0.9038)

Live Demonstration Master Cue Sheet

Speaker	Duration	PPT Slides Covered	Live Streamlit App Modules Demonstrated
1. Dhruv	2.5 min	Slide 1 (Title & Team) Slide 2 (Problem & Objectives) Slide 3 (Dataset & Quality)	• Module 1: Overview & Mission Control • Module 2: Train Delay Predictor (Preset 1 & 2)
2. Darshi	2.5 min	Slide 4 (EDA & Key Insights) Slide 5 (Preprocessing Pipeline) Slide 6 (Feature Engineering)	• Module 4: Exploratory Data Analysis (EDA) • Module 5: Data Preprocessing & Pipeline • Module 6: Feature Engineering & Selection
3. Kirtan	3.0 min	Slide 7 (Workflow & Baseline) Slide 8 (Tuning & Algorithms) Slide 9 (Comparison Table) Slide 10 (Confusion Matrix) Slide 11 (Error Analysis) Slide 12 (Conclusion & Q&A;)	• Module 7: Model Development & Benchmarks • Module 8: Evaluation Metrics & Confusion Matrix • Module 9: Error Analysis & Misclassifications • Module 10: Model Interpretation (Feature Imp.)

■■ SPEAKER 1: DHRUV — INTRODUCTION & PROBLEM FORMULATION

■ SHOW PPT SLIDE 1 & SLIDE 2 (Title, Team, Problem & Objectives)

*“Good morning respected faculty members, evaluators, and colleagues. My name is **Dhruv**, and along with my team members **Darshi** and **Kirtan**, we present our Machine Learning project: **'Indian Railways Train Delay Prediction and Network Punctuality Analytics'**.*

*Indian Railways operates one of the busiest rail networks globally, carrying over 23 million passengers daily. However, reactive delay management leads to cascading congestion across junction points. Our objective is to formulate a proactive binary classification model predicting whether a train will experience an arrival delay exceeding **15 minutes** at its final terminus (labeled 1 for Delayed, and 0 for On-Time), adhering to official Ministry of Railways punctuality standards.”*

■■ SPEAKER 1: DHRUV — DATASET OVERVIEW & TARGET LEAKAGE QUARANTINE

■ SHOW PPT SLIDE 3 (Dataset Overview & Quality Assessment)

*“Our analysis is based on 10,000 operational records sampled from historical Indian Railways logs across 45 raw features. The empirical target distribution is naturally imbalanced: **72.16% delayed versus 27.84% on-time**.*

Crucially, we identified post-journey features—such as delay_minutes and primary_delay_cause—which represent direct target leakage. We strictly quarantined these columns so our model only uses information available prior to train dispatch. Now, let me demonstrate our live prediction dashboard.”

■■ SPEAKER 1: DHRUV — LIVE STREAMLIT APP PREDICTOR DEMONSTRATION

■ SWITCH TO LIVE WEB APP: Module 1 (Overview) & Module 2 (Delay Predictor)

*“Here on screen is our live Streamlit application featuring an enterprise Deep Dark and Elegant Gold UI. On the Overview dashboard, operators monitor network-wide KPIs. Let us navigate to the **Train Delay Predictor** to test two scenarios.”*

■ [ACTION 1] Select Preset 1: Premier Vande Bharat Express (NR - Delhi to Varanasi, 130 km/h, clear weather, LHB rake).

■ [CLICK] 'Run ML Delay Prediction' -> Result: 3.3% Delay Probability -> GREEN: ON-TIME EXPECTED.

■ [ACTION 2] Select Preset 2: Howrah-Amritsar Mail in Heavy Monsoon (ER - single track, 24 stops, 88% congestion, late rake).

■ [CLICK] 'Run ML Delay Prediction' -> Result: 98.7% Delay Probability -> RED ALERT: HIGH DELAY RISK.

■ [HANDOVER] 'I now hand over to Darshi to explain our data preprocessing and feature engineering.'

■■ SPEAKER 2: DARSHI — EXPLORATORY DATA ANALYSIS (EDA)

■ SHOW PPT SLIDE 4 (Exploratory Data Analysis & Key Insights)

*"Thank you, Dhruv. Good morning everyone, I am **Darshi**.*

During Exploratory Data Analysis, we identified three vital operational patterns:
*1. **Kinematic Threshold:** Trains averaging below 50 km/h suffered an 88.4% delay rate, while trains exceeding 75 km/h maintained under 15% delay incidence.*
*2. **Zonal Network Friction:** Northern and Eastern railway zones suffer the highest congestion (>0.85), severely impacted by winter fog.*
*3. **Cascading Turnaround Impact:** When a shared incoming rake is delayed, downstream journey delay probability surges to 91.2%, proving that delays cascade across schedules."*

■■ SPEAKER 2: DARSHI — DATA PREPROCESSING & NUMPY IQR WINSORIZATION

■ SHOW PPT SLIDE 5 (Data Preprocessing & Pipeline Architecture)

"To clean and prepare the data without introducing bias, we implemented a 4-stage pipeline:
• Constant zero-variance features like is_overloaded were eliminated.
*• Rather than discarding extreme journeys, we applied robust **NumPy IQR Winsorization** strictly fitted on training data: capping distance_km at **1,466 km** and scheduled_travel_hours at **24.9 hours**.*
• High-cardinality categoricals were one-hot encoded into 8,111 aligned sparse columns, and numerical features were standardized using training-set Z-scores."

■■ SPEAKER 2: DARSHI — FEATURE ENGINEERING & LIVE DATA DEMONSTRATION

■ SHOW PPT SLIDE 6 & SWITCH TO LIVE WEB APP (Module 4, 5, 6)

"We engineered five high-impact railway domain features: avg_speed_kmh, network_vulnerability, equipment_age_index, rake_turnaround_risk, and trigonometric cyclical time encodings (hour_sin and hour_cos)."

■ [SHOW LIVE APP Module 4: EDA] Display target distribution (72/28 split) and speed-delay correlation plots.

■ [SHOW LIVE APP Module 5: Preprocessing] Highlight the 4-stage pipeline diagram and verified IQR clipping thresholds.

■ [SHOW LIVE APP Module 6: Feature Engineering] Walk through the mathematical formulas for network vulnerability and equipment age.

■ [HANDOVER] 'I now pass the floor to Kirtan, who will detail our model benchmarks, LightGBM tuning, and error diagnostics.'

■■ SPEAKER 3: KIRTAN — WORKFLOW, BASELINE & ALGORITHM BENCHMARKS

■ SHOW PPT SLIDE 7, 8, 9 (Machine Learning Workflow, Tuning & Comparison)

*"Thank you, Darshi. Good morning respected evaluators, I am **Kirtan**.*

*To establish a baseline, we implemented a **Dummy Majority-Class Classifier**. Because 72.15% of journeys are delayed, the baseline achieves 72.15% accuracy and 100% recall by predicting 'Delayed' every time. However, its ROC-AUC is exactly **0.5000** and Precision is only 72.15%—it has zero discriminative intelligence.*

We evaluated four diverse algorithms: Decision Trees, Random Forests, L2-Regularized Logistic Regression, and LightGBM. On our untouched validation set of 2,000 journeys:
• Decision Tree: 81.30% Accuracy | 0.8593 ROC-AUC
• Random Forest: 78.70% Accuracy | 0.8854 ROC-AUC
*• Logistic Regression: 85.70% Accuracy | **0.9229 ROC-AUC***
*• **Tuned LightGBM (Champion): 85.85% Accuracy | 88.72% Precision | 92.10% Recall | 0.9038 F1-Score | 0.9121 ROC-AUC**"*

■■ SPEAKER 3: KIRTAN — CONFUSION MATRIX & ERROR ANALYSIS

■ SHOW PPT SLIDE 10 & 11 (Confusion Matrix, Error Analysis & Limitations)

*"In passenger railway dispatching, a **False Negative** (failing to detect a delayed train) causes severe platform gridlock, whereas a **False Positive** merely triggers precautionary monitoring. LightGBM achieves a **92.10% Recall**, successfully catching over 9 out of 10 delayed trains.*

At threshold 0.50 across 2,000 journeys: True Positives = 1,329, True Negatives = 388, False Positives = 169 (8.45%), and False Negatives = 114 (5.70%). False Negatives primarily occurred during sudden unscheduled weather and mechanical failures that cannot be anticipated from pre-departure schedules alone."

■■ SPEAKER 3: KIRTAN — LIVE UI EVALUATION DEMO & CONCLUSION

■ SWITCH TO LIVE WEB APP (Module 7, 8, 9, 10) & SHOW PPT SLIDE 12

"Let me demonstrate these evaluation tools in our web platform:"

■ [SHOW LIVE APP Module 7: Benchmarks] Point out the validation bar chart comparing Accuracy, Recall, and F1 across all models.

■ [SHOW LIVE APP Module 8: Evaluation] Demonstrate the interactive threshold explorer slider (adjust from 0.50 down to 0.35 to push recall above 96%).

■ [SHOW LIVE APP Module 10: Feature Importance] Highlight top tree-split drivers: avg_speed_kmh (570), network_vulnerability (309), and equipment_age_index (221).

■ [CONCLUSION] 'To summarize: our team has developed a production-ready, leakage-free ML pipeline with 85.85% accuracy and 92.10% recall, paired with a tested 13-module operations suite. On behalf of Dhruv, Darshi, and myself, thank you. We welcome your questions.'

Essential Viva Defense Cheat-Sheet

Concept	Key Project Defense & Numbers
Target Definition	Binary classification of is_delayed (>15 minutes arrival delay at destination terminus).
Data Leakage Defense	Quarantined delay_minutes and primary_delay_cause because they are recorded post-journey.
Outlier Strategy	NumPy IQR Winsorization on training set: capped distance at 1,466 km, hours at 24.9 hrs. Zero data rows deleted.
Champion Model	Tuned LightGBM (85.85% Acc, 88.72% Prec, 92.10% Rec, 0.9038 F1, 0.9121 AUC). Highest balanced F1-score.
Confusion Matrix (n=2,000)	TP = 1,329 | TN = 388 | FP = 169 (8.45%) | FN = 114 (5.70%). High recall minimizes costly missed delays.